

Bench sheet: steps 1 and 2

Convert the clock, flash the merged firmware, then measure the string. Rev 0.4 amended, 20 Sep 2026.

Before you start

No cells connected for any of this. Use a current-limited bench supply, 200 mA limit, for VCC.
Do not connect the panel or the charger. R3 sits at the board end, not out at the string.
Duty does not limit current: only R3 and the pin do. Verify at 4.20 V and 100% duty.

1 Convert the ATmega328P to internal 8 MHz, BOD 2.7 V (closes H03)

Power the board from the ISP or a bench supply, never both. Battery and panel disconnected.

Fuse	Value	Meaning
Low	0xE2	Internal 8 MHz RC, CKDIV8 off, slow rising power
High	0xD9	Factory default: SPIEN on, no boot reset vector
Extended	0xFD	BOD 2.7 V. avrdude may read it back as 0x05: same three bits

avrdude -c usbasp -p m328p -U lfuse:w:0xE2:m -U hfuse:w:0xD9:m -U efuse:w:0xFD:m
Arduino as ISP instead: -c stk500v1 -P /dev/cu.usbmodemXXXX -b 19200
Or MiniCore: ATmega328P, Clock "Internal 8 MHz", BOD 2.7 V, Bootloader "No bootloader", Burn Bootloader.

Read back and confirm: avrdude -c usbasp -p m328p -U lfuse:r:-:h -U hfuse:r:-:h -U efuse:r:-:h

Chip erase removes any bootloader. From here on, upload with the programmer, not over serial.

Board work: remove the regulator and the power LED or its series resistor. Feed VCC directly, leave RAW unused, keep the local decoupling. The 16 MHz crystal can stay fitted; it is simply not selected.

Pass: fuses read back as written, and the board runs with VCC at 3.3 V.

2 Flash the merged firmware and check it thinks straight

Build with DEBUG_SERIAL 1 and TIME_SCALE 60, F_CPU 8 MHz. Upload with the programmer. Serial at 9600.

Check	Expect
Serial line per wake	t=... vdd=3.80 state=... pct=...
VCC at 3.80 V	vdd within 0.05 V of the meter (set BANDGAP_V = 1.10 x Vmeter / Vreported)
D9 PWM frequency	1.9 to 2.0 kHz, confirming the 8 MHz clock
Cover the LDR 5 s	state goes to night, pct ramps to 100 (5 s = 5 "minutes" at TIME_SCALE 60)
Uncover the LDR	state returns to day, pct fades to 0

3 Gate 5: string current with R3 = 47 ohm (settles 47 vs 56 ohm)

DMM in series with the string positive. Lights at 100% duty (LDR covered). Record what you read:

Supply	Expect	Measured	R3 temp	Notes
3.40 V	11 mA			
3.70 V	15 mA			
4.20 V	21 mA			

Decision rule

Over 22 mA at 4.20 V: fit 56 ohm and repeat the three readings (expect 18 / 13 / 9 mA).
Within tolerance but too dim from the street at 3 am: keep 47 ohm and accept the 21 mA peak,
or cap commanded duty above 4.0 V in firmware. Tell Claude which and it changes three files.

Also confirm: R3 and the D9 pin stay at ambient; the string is dark with M2 unplugged and during reset.

Then restore DEBUG_SERIAL 0 and TIME_SCALE 1 before any real-time LVC or overnight test.